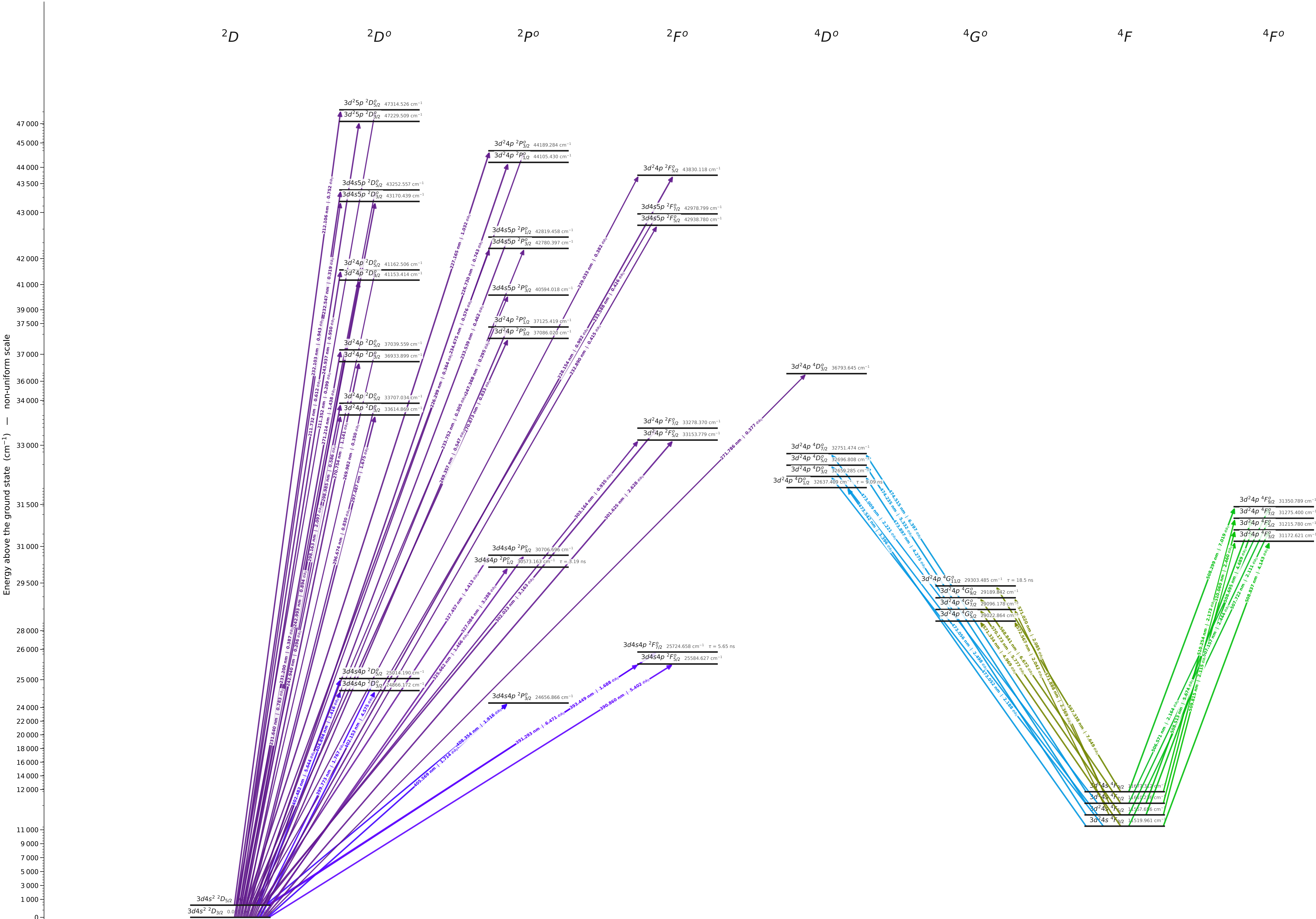
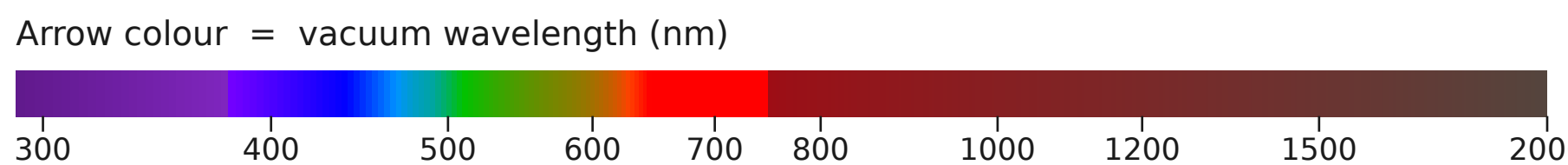


# Scandium

Sc (Sc II). Strongest classified transitions below 2 μm (NIST): 49 levels, 70 lines.

Sc<sup>+</sup> ionisation limit 52922.00 cm<sup>-1</sup> = 6.56149 eV (188.957 nm)

Arrow label:  $\lambda$  vacuum wavelength | reduced dipole moment element  $\langle U||e\Gamma||U\rangle$  in  $e a_0$   
Matrix elements are measured or NIST values: \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: NIST energy and measured lifetime (where one exists).



## Key transitions

| Transition                                      | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ \langle U  e\Gamma  U\rangle $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use           |
|-------------------------------------------------|--------------------|--------------------|-------------|-----------------------------------------------------|----------------------|-------------------------|------------|---------------|
| $3d^24s\ ^4F_{9/2} - 3d^24p\ ^4G^{\circ}_{5/2}$ | 567.3381           | 567.1807           | 528.4194    | 7.649                                               | 5.410e+07            | 8.612                   | 100        | laser-cooling |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level, 1/(2 $\pi\tau$ ).

## <sup>45</sup>Sc hyperfine structure ( $I = 7/2$ )

| Level                        | A (MHz)      | B (MHz)     | Shift of each F level from the centre of gravity (MHz)                                                         |
|------------------------------|--------------|-------------|----------------------------------------------------------------------------------------------------------------|
| $3d4s^2\ ^2D_{3/2}$          | 269.5581(9)  | 26.361(9)   | F=2: -1805.39 F=3: -1015.55 F=4: +55.15 F=5: +1421.77                                                          |
| $3d4s^2\ ^2D_{5/2}$          | 109.0330(9)  | -37.372(15) | F=1: -1246.64 F=2: -1017.90 F=3: -678.79 F=4: -234.11 F=5: +309.72 F=6: +944.70                                |
| $3d^24s\ ^4F_{3/2}$          | -158.512(3)  | -5.189(21)  | F=2: +1067.18 F=3: +595.35 F=4: -37.22 F=5: -833.49                                                            |
| $3d^24s\ ^4F_{5/2}$          | 154.0352(21) | -6.45(3)    | F=1: -1736.35 F=2: -1426.44 F=3: -962.26 F=4: -344.64 F=5: +425.30 F=6: +1346.20                               |
| $3d^24s\ ^4F_{7/2}$          | 250.0119(21) | -9.08(3)    | F=0: -3942.55 F=1: -3691.61 F=2: -3189.92 F=3: -2437.85 F=4: -1435.95 F=5: -184.96 F=6: +1314.18 F=7: +3060.38 |
| $3d^24s\ ^4F_{9/2}$          | 285.966(9)   | -15.5(7)    | F=1: -5511.94 F=2: -4937.79 F=3: -4077.04 F=4: -2930.23 F=5: -1498.10 F=6: +218.43 F=7: +2218.26 F=8: +4500.10 |
| $3d4s4p\ ^2P^{\circ}_{1/2}$  | 442(5)       | -           | F=3: -994.26 F=4: +773.32                                                                                      |
| $3d4s4p\ ^2P^{\circ}_{3/2}$  | 12.6(9)      | -191(13)    | F=2: -187.13 F=3: -13.17 F=4: +91.67 F=5: +18.44                                                               |
| $3d4s4p\ ^2D^{\circ}_{3/2}$  | 278(5)       | -           | F=2: -1875.88 F=3: -1042.15 F=4: +69.48 F=5: +1459.02                                                          |
| $3d4s4p\ ^2D^{\circ}_{5/2}$  | 31(4)        | -           | F=1: -347.38 F=2: -285.63 F=3: -192.99 F=4: -69.48 F=5: +84.92 F=6: +270.19                                    |
| $3d4s4p\ ^2F^{\circ}_{5/2}$  | 171.5(15)    | -           | F=1: -1929.16 F=2: -1586.20 F=3: -1071.76 F=4: -385.83 F=5: +471.57 F=6: +1500.46                              |
| $3d4s4p\ ^2F^{\circ}_{7/2}$  | 125(3)       | -           | F=0: -1964.24 F=1: -1839.53 F=2: -1590.10 F=3: -1215.96 F=4: -717.10 F=5: -93.53 F=6: +654.75 F=7: +1527.74    |
| $3d^24p\ ^4G^{\circ}_{5/2}$  | 279.1(12)    | -29(7)      | F=1: -3155.69 F=2: -2589.08 F=3: -1742.32 F=4: -619.18 F=5: +775.31 F=6: +2434.84                              |
| $3d^24p\ ^4G^{\circ}_{7/2}$  | 147.2(3)     | -           | F=0: -2318.37 F=1: -2171.17 F=2: -1876.78 F=3: -1435.18 F=4: -846.39 F=5: -110.40 F=6: +772.79 F=7: +1803.18   |
| $3d^24p\ ^4G^{\circ}_{9/2}$  | 83.3(6)      | -25(4)      | F=1: -1615.74 F=2: -1445.51 F=3: -1190.89 F=4: -852.78 F=5: -432.36 F=6: +68.88 F=7: +649.16 F=8: +1306.42     |
| $3d^24p\ ^4G^{\circ}_{11/2}$ | 44.7(3)      | -39(7)      | F=2: -1032.47 F=3: -892.03 F=4: -706.22 F=5: -476.24 F=6: -203.63 F=7: +109.76 F=8: +461.81 F=9: +850.06       |
| $3d4s4p\ ^2P^{\circ}_{1/2}$  | 240(19)      | -           | F=3: -539.63 F=4: +419.71                                                                                      |
| $3d4s4p\ ^2P^{\circ}_{3/2}$  | 327(4)       | -           | F=2: -2209.77 F=3: -1227.65 F=4: +81.84 F=5: +1718.71                                                          |
| $3d^24p\ ^4F^{\circ}_{5/2}$  | 280.3(3)     | -           | F=2: -1892.07 F=3: -1051.15 F=4: +70.08 F=5: +1471.61                                                          |
| $3d^24p\ ^4F^{\circ}_{7/2}$  | 121.1(3)     | -           | F=1: -1362.56 F=2: -1120.32 F=3: -756.98 F=4: -272.51 F=5: +333.07 F=6: +1059.77                               |
| $3d^24p\ ^4F^{\circ}_{9/2}$  | 67.5(3)      | -           | F=0: -1062.39 F=1: -994.94 F=2: -860.03 F=3: -657.67 F=4: -387.86 F=5: -50.59 F=6: +354.13 F=7: +826.30        |
| $3d^24p\ ^4F^{\circ}_{11/2}$ | 44.4(3)      | -           | F=1: -854.11 F=2: -765.37 F=3: -632.26 F=4: -454.79 F=5: -232.94 F=6: +33.28 F=7: +343.86 F=8: +698.82         |
| $3d^24p\ ^4D^{\circ}_{5/2}$  | 432.9(9)     | -           | F=3: -974.03 F=4: +757.58                                                                                      |
| $3d^24p\ ^4D^{\circ}_{7/2}$  | 126.2(15)    | -           | F=2: -851.93 F=3: -473.30 F=4: +31.55 F=5: +662.62                                                             |
| $3d^24p\ ^4D^{\circ}_{9/2}$  | 70.5(9)      | -           | F=1: -792.58 F=2: -651.67 F=3: -440.32 F=4: -158.51 F=5: +193.74 F=6: +616.45                                  |
| $3d^24p\ ^4D^{\circ}_{11/2}$ | 42.0(3)      | -           | F=0: -661.04 F=1: -619.07 F=2: -535.13 F=3: -409.22 F=4: -241.33 F=5: -31.48 F=6: +220.35 F=7: +514.14         |
| $3d^24p\ ^4F^{\circ}_{5/2}$  | 226.9(18)    | -           | F=1: -2553.11 F=2: -2099.22 F=3: -1418.39 F=4: -510.62 F=5: +624.09 F=6: +1985.75                              |
| $3d^24p\ ^4F^{\circ}_{7/2}$  | 39.9(6)      | -           | F=0: -627.99 F=1: -588.12 F=2: -508.37 F=3: -388.76 F=4: -229.27 F=5: -29.90 F=6: +209.33 F=7: +488.44         |
| $3d^24p\ ^4D^{\circ}_{5/2}$  | 236(3)       | -           | F=2: -1590.55 F=3: -883.64 F=4: +58.91 F=5: +1237.09                                                           |
| $3d^24p\ ^4D^{\circ}_{7/2}$  | 41.7(15)     | -           | F=1: -468.80 F=2: -385.46 F=3: -260.44 F=4: -93.76 F=5: +114.60 F=6: +364.62                                   |
| $3d^24p\ ^4D^{\circ}_{9/2}$  | 56(4)        | -           | F=2: -376.39 F=3: -209.10 F=4: +13.94 F=5: +292.75                                                             |
| $3d^24p\ ^4D^{\circ}_{11/2}$ | 162.2(15)    | -           | F=2: -1094.77 F=3: -608.20 F=4: +40.55 F=5: +851.49                                                            |
| $3d^24p\ ^4D^{\circ}_{13/2}$ | 133.1(9)     | -           | F=1: -1497.46 F=2: -1231.25 F=3: -831.92 F=4: -299.49 F=5: +366.05 F=6: +1164.69                               |
| $3d^24p\ ^4P^{\circ}_{5/2}$  | 215.6(9)     | -           | F=2: -1454.97 F=3: -808.32 F=4: +53.89 F=5: +1131.64                                                           |
| $3d^24p\ ^4P^{\circ}_{7/2}$  | 110.3(15)    | -           | F=3: -248.23 F=4: +193.07                                                                                      |
| $3d4s5p\ ^2P^{\circ}_{3/2}$  | 136(4)       | -           | F=2: -918.71 F=3: -510.40 F=4: +34.03 F=5: +714.55                                                             |
| $3d^24p\ ^4D^{\circ}_{5/2}$  | -195(4)      | -           | F=2: +1317.36 F=3: +731.87 F=4: -48.79 F=5: -1024.62                                                           |
| $3d^24p\ ^4D^{\circ}_{7/2}$  | 469.2(18)    | -           | F=1: -5278.22 F=2: -4339.87 F=3: -2932.34 F=4: -1055.64 F=5: +1290.23 F=6: +4105.28                            |
| $3d4s5p\ ^2P^{\circ}_{1/2}$  | 758(9)       | -           | F=2: -5117.68 F=3: -2843.16 F=4: +189.54 F=5: +3980.42                                                         |
| $3d4s5p\ ^2F^{\circ}_{5/2}$  | 67(27)       | -           | F=0: -1057.67 F=1: -990.51 F=2: -856.21 F=3: -654.75 F=4: -386.13 F=5: -50.37 F=6: +352.56 F=7: +822.63        |
| $3d4s5p\ ^2D^{\circ}_{5/2}$  | 342(4)       | -           | F=2: -2310.95 F=3: -1283.86 F=4: +85.59 F=5: +1797.41                                                          |
| $3d4s5p\ ^2D^{\circ}_{7/2}$  | 219(11)      | -           | F=1: -2468.79 F=2: -2029.89 F=3: -1371.55 F=4: -493.76 F=5: +603.48 F=6: +1920.17                              |
| $3d^24p\ ^2F^{\circ}_{5/2}$  | 11(5)        | -           | F=1: -128.16 F=2: -105.38 F=3: -71.20 F=4: -25.63 F=5: +31.33 F=6: +99.68                                      |
| $3d^24p\ ^2P^{\circ}_{1/2}$  | 463(28)      | -           | F=3: -1040.80 F=4: +809.51                                                                                     |

Measured A, B constants; sources in the data file.